INVESTIGATING INSTRUCTIONAL DESIGN FOR GPT PROMPT TRAINING: A PILOT STUDY ON USER LEARNING, SATISFACTION, AND AI-ASSISTED LABELING IMPACT

Dustin Tran[1], Qile Wang[2], Matthew Louis Mauriello[3]

1: Undergraduate Student, University of Delaware, dustintr@udel.edu
2: Graduate Student, University of Delaware, kylewang@udel.edu
3: Assistant Professor, University of Delaware, mlm@udel.edu

INTRODUCTION

This study builds on our previous work, "The explanation makes sense: An empirical study on LLM performance in news classification and its influence on judgment in human-AI collaborative annotation," and the SMIDGen platform. Our prior findings showed that **well-crafted prompts significantly improved multi-class classification accuracy**. With this in mind, we explore how SMIDGen's onboarding process can move beyond traditional instructional design toward a more **personalized approach**. Instead of a one-size-fits-all training, we propose using GPT-driven feedback to tailor prompt-writing guidance to individual users. While our study is still ongoing, we predict that users who receive personalized training will perform better in AI-Human collaborated annotating tasks.

OBJECTIVE

Our study investigates the most effective instructional design approaches for learning prompt engineering by examining two key factors: (1) **task complexity** and (2) **the level of feedback** provided after users enter their prompts. Specifically, we aim to determine whether personalized AI feedback leads to **measurable improvements** compared to traditional human-written guidelines or no feedback at all. Additionally, we want to build the foundations for future study to explore the potential for AI-generated suggestions to either **enhance user learning or encourage dependency**, particularly in tasks requiring significant **social context and subjectivity**. Since LLMs are inherently influenced by the biases in their training data, we emphasize the importance of human oversight in prompt engineering. While our previous study demonstrated that explainable AI fosters trust in human-AI collaboration, our goal now is to identify instructional strategies that balance AI assistance with human expertise, ensuring users develop strong, adaptable prompt-writing skills.

LITERATURE REVIEW

Prompt Engineering Training on Journalists

This research highlights key considerations for our study. *Bashardoust et al. (2024)* found that the **effectiveness of prompt engineering training is highly task-dependent**, as accuracy improvements varied based on task difficulty. This suggests that training is not universally beneficial across all tasks, reinforcing our decision to limit our survey study to a specific dataset that aligns well with few-shot prompting techniques.

Furthermore, the study observed that a holistic training session ***increased* journalists' perceived expertise in LLM use**. However, it simultaneously ***decreased* their perception of LLM helpfulness**, likely due to greater awareness of LLM limitations. This aligns with our concern that users could become overly reliant on LLMs for classification tasks. As a result, we emphasize not only training users in effective prompting but also educating them on the **limitations of LLMs to encourage critical evaluation of model outputs**.

Few-shots Prompt Training

Prabhumoye et al. (2022) demonstrate that few-shot instruction-based training can be **effectively applied to binary and multi-class tasks**, avoiding the need for large labeled datasets and fine-tuning. Inspired by this, we implement few-shot prompting in our study as a practical method for novice users to practice prompt engineering, given our limited dataset size. We believe that practicing with few-shot prompting will better prepare users to adapt this technique to their own datasets in the future, compared to simpler techniques like zero-shot learning.



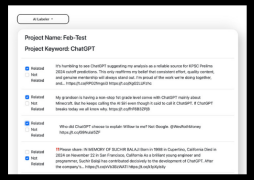
PROMPT LEARNING FOR AI-HUMAN COLLABORATION IN PROGRAMMATIC LABELING:

ENHANCING *NOT REPLACING*



Image was found on Pinterest

BACKGROUND



We are developing SMIDGen, a platform designed to help researchers collect, analyze, and label data more efficiently. A key feature, the AI Labeler, enhances the labeling process through human-AI collaboration, particularly for large datasets. To optimize this tool, we are investigating the best instructional designs for training researchers to write effective prompts that improve AI-assisted labeling.

Building on our previous study, which explored how GPT-generated explanations influenced human decision-making in the context of news classification, we found that explainable AI can significantly increase self-reported confidence and alter decisions. Participants were more likely to align with GPT's conclusions after reviewing its explanations, though some were misled by incorrect information. This highlights the need to balance the benefits of AI-generated explanations with the risk of over-reliance on AI. This is why we aim to ensure that researchers achieve effective prompt-writing results without becoming dependent on AI suggestions. This will help them maintain control over their work, especially when handling subjective or context-rich datasets.

SENSIFY LAB · SMID + GEN+ · UNIVERSITY OF DELAWARE

SURVEY DESIGN

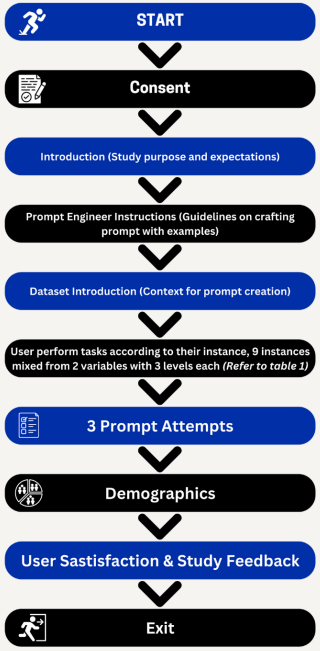


Figure 2: Survey Flow Graph

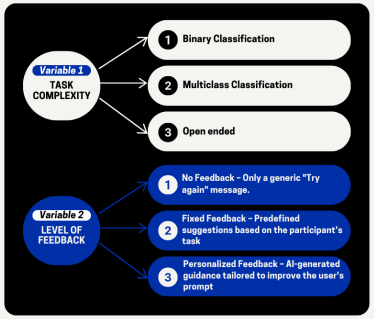


Figure 1: Variables of the study

Instance	Variable 1	Variable 2
1	Binary Classification	No feedback
2	Binary Classfication	Fixed feedback
3	Binary Classification	Personalized feedback
4	Multiclass Classification	No feedback
5	Multiclass Classification	Fixed feedback
6	Multiclass Classification	Personalized feedback
7	Open ended	No feedback
8	Open ended	Fixed feedback
9	Open ended	Personalized feedback

Table 1: Instances of the Survey for In-between subjects



Figure 3: Example of a qualtric prompt

EXPECTED OUTCOMES

- We expect participants to show significant improvement in performance when receiving personalized feedback generated by the AI compared to those who receive no feedback or fixed, human-written feedback.
- We anticipate that users will become more dependent on the feedback provided by the LLM as they progress through the trials. This dependence may increase over time, especially for participants who experience notable improvement when using AI-generated feedback.
- We expect a higher level of user satisfaction with the AI-generated personalized feedback compared to fixed, human-written feedback.

REFERENCES

Bashardoust, Amirsiavosh, et al. "The Effect of Education in Prompt Engineering: Evidence from Journalists." arXiv preprint arXiv:2409.12320 (2024).

Prabhumoye, Shrimai, et al. "Few-shot instruction prompts for pretrained language models to detect social biases." arXiv preprint arXiv:2112.07868 (2021).

Fagbohun, O., Harrison, R. M., & Dereventsov, A. An empirical categorization of prompting techniques for large language models: A practitioner's guide, 2024. URL https://arxiv.org/abs/2402. 14837v1.

"Integrating/Embedding ChatGPT for Multiple Turns." Qualtrics Community, 10 Feb. 2023, https://community.qualtrics.com/custom%2Dcode%2D12/integrating%2Dembedding%2Dchat%2Dgpt%2Dfor%2Dmultiple%2Dturns%2D26923.